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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	INTELLIGENT SUPPLIER RISK PREDICTION AND ANALYTICS	
S. No.	University ID	Name
1	2320030092	GULLAPALLI SHEETAL
2	2320030098	JANGARI SAKSHITHA
3	2320030102	GULLAPALLI NIKITHA CHOWDARY
Name of the Guide		Dr N Shirisha

Abstract

Modern supply chains depend heavily on suppliers for raw materials, components, and essential services. Supplier disruptions caused by financial instability, operational problems, production delays, negative business news, or unexpected external events can significantly affect an organization's production and business operations. Traditional supplier evaluation methods mainly depend on historical financial reports and periodic assessments, which may not identify rapidly developing risks at an early stage. This project proposes an AI-based Supplier Risk Prediction System that combines large-scale financial and news data to predict potential supplier disruptions up to 90 days in advance.

The proposed system uses Azure for cloud-based data storage and processing. Apache Spark with PySpark is used to efficiently clean, transform, and process large volumes of supplier financial and news data. Financial factors such as revenue, debt, leverage, current ratio, and financial stability are analyzed using XGBoost to estimate financial risk. Supplier-related financial news is processed using FinBERT to identify positive, negative, and neutral sentiment. The resulting sentiment information is organized into a 90-day sequence and analyzed using an LSTM deep learning model to identify changing risk patterns over time. The outputs from the financial and news models are combined to generate an overall supplier disruption risk score and classify suppliers into low, medium, and high-risk categories.

The processed data and prediction results are connected to Power BI to create an interactive supplier risk dashboard. The dashboard provides visual insights into supplier risk levels, financial indicators, news sentiment, and risk trends. The proposed system helps organizations identify vulnerable suppliers early, take preventive measures, consider alternative suppliers, and improve supply chain resilience and decision-making.

Signature of the Guide

HOD